Component	Voltage (V)	Current (mA)	Power (mW)	Source / Notes
PIC18F57Q43 Curiosity Nano	5	50	250	**Conservative design assumption**
NAU7802KGI	5	2.1	10.5	Datasheet operating current
1 kg Load Cell	5	15	75	Conservative design assumption
Status LED + R3	5	1.96	9.8	Calculated: (5 V - 2.65 V) / 1.2 kOhm
Tare/Zero Pull-up R4	5	0.5	2.5	Worst case with button pressed
R5/R6 Digital Pull-ups	5	2.13	10.65	Worst-case combined
Curiosity Nano External-Power Overhead	5	0.1	0.5	USB connected; board overhead
R7/R8 Command Pull-downs	5	0.1	0.5	R7/R8 100 kOhm pull-downs; worst-case combined
ESTIMATED TOTAL		71.89	359.45	
25% DESIGN MARGIN		17.97		
DESIGN REQUIREMENT		89.86		
FUSE LIMIT		500		
REMAINING CURRENT MARGIN		410.14		

LM7805 Thermal Budget

Condition	**Input Voltage (V)**	**Output Voltage (V)**	**Design Current (A)**	**Regulator Dissipation (W)**
Minimum Input	9	5	0.09	0.36
Maximum Input	12	5	0.09	0.63